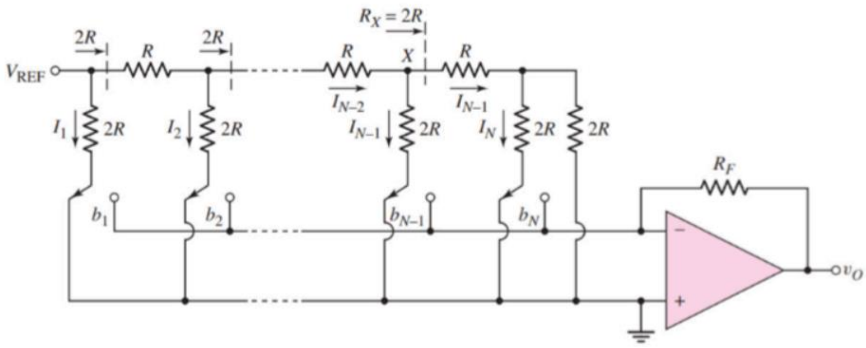
CSE 350

Digital Electronics and Pulse Techniques

Assignment-01

Total – 20 marks

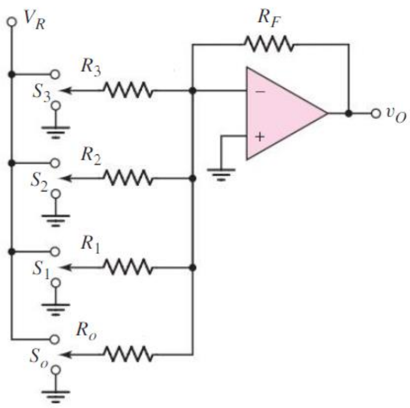
Question 1



The N-bit DAC with an R-2R ladder network in the above figure is to be designed as a 3-bit DAC device. Suppose V_{REF} = -10V, R_F = 10 KΩ and R = 5KΩ.

(a)	Determine the currents I_1, I_2, and I_3.	1
(b)	Determine the resolution of the circuit.	1
(c)	Determine the maximum and minimum value of Vo.	1
(d)	Determine the ½ MSB value.	1
(e)	Draw the Vin Vs. Vout graph with proper labeling.	1

Question 2



Here, VR=-8V, R3=Rf & Input bit sequence: S3 S2 S1 S0

(a)	Show the output voltages for all possible input cases 'XXX0' in a table, where X may be 0 or 1.	2
(b)	What is the ratio of maximum output voltage & LSB (for input 0001) voltage?	1
(c)	What should be the ratio R3Rf, if we want the output voltage for input '1010' to be 5V?	2

Question 3

The figure shows a 2-bit Flash ADC with V_{REF} = 10V and its input ranged from 0 to 10V. Given: R1 = 5kΩ, R2 = 10kΩ, R3 = 10kΩ, R4 = 15kΩ. The two outputs of the ADC are connected to two LEDs.

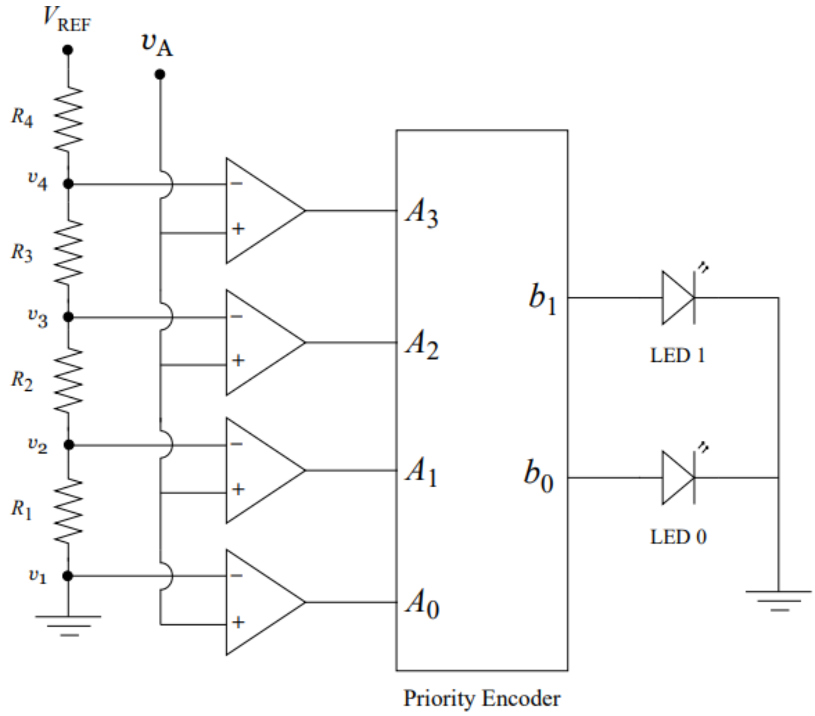


Figure 5: 2-bit flash ADC

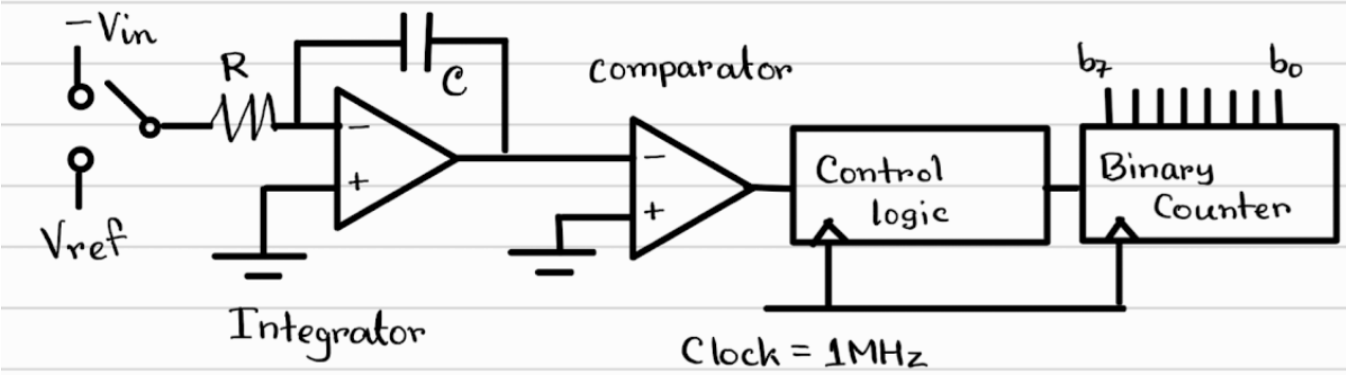
Sampled Value (V)	Time (s)
5.00	0.0
7.40	0.5
5.71	3.0
1.22	4.0

Table 1: Sampled values at different time instances

(a)	Determine the quantization range, corresponding digital output, and the states of the LED for the 2-bit Flash ADC shown in Figure 5.	2.5
(b)	Four sample values at four different time instances obtained from an analog signal are shown in Table 1. Identify the quantization range for each sampled value. Determine the corresponding encoded output, and identify the states of the LEDs.	2.5

Question 4

A dual slope ADC has Vref = 5V, and a 8 bit counter that outputs an 8 bit representation of the input signal. A 1 MHz clock is used for the clock and counter circuits.



(a)	Determine the input voltage range, number of steps, and step size of the ADC.	1
(b)	For a specific input voltage, the ADC outputs a count of n=200. Determine the input voltage.	1
(c)	Determine the total reading time or conversion time in (b).	1
(d)	Calculate the maximum sampling rate.	1
(e)	If we wish to get a sampling rate 4 times that of the current circuit, determine how many bits should be used for the counter without changing any other system parameter.	1